

GREEN TECHNOLOGY

"Green generators are noiseless. These cut off the dependence on exhaustive fuels, utilising renewable energy instead. These also avoid environmental pollution caused by diesel generator sets.

"Properly-designed solar generators can run 24/7. These can store sunlight energy through the day, while quick charging lithium-ion batteries through grid electricity at night whenever the charge falls below 50 per cent. Overall, these generators can bring the cost down by 50 to 95 per cent, considering expenses on fuel, electricity and maintenance."

Many solar generators are portable and feature plug-and-play characteristics. This means these can be set up on wheels and installed anywhere suitable. These generators start functioning almost instantly after setup.

The setup

Solar-powered generators function through a number of units. To start, solar panels of desired capacity are positioned and connected to a battery. Lithium-ion batteries are currently the most preferred choice in the market for their various advantages, even though these are costlier than the others (like lead-acid batteries).

The battery unit is charged by the energy created by the solar panels by capturing sunlight. A charge controller is connected to the battery unit to optimise storage. An inverter converts direct current (DC) supplied from the battery storage unit into alternating current (AC) to power appliances. A fuel tank is accommodated as backup in case the battery fails to operate.

Most solar generators allow customisations. Solar panels, battery capacities, inverters and more can be modified as per requirement.

Cost and returns

Solar-powered generators are expensive in terms of upfront investment. These may cost about 10 times more than regular ones. To explain this, Patel uses the example of green generators, manufactured by Netherlands-based Maru Systems International and distributed in India by Divya Hitech. He says, "A 44kVA generator set with 600Ah - 8000Ah lithium battery capacity and 1kW-peak (kWp) solar panel costs approximately ₹ 8 million (including GST). The battery itself costs about ₹ 800,000." However, drastic reduction in running expenses ensures a quick return on investment.

Patel adds, "To give you an idea, a 30kVA traditional generator, running 24/7, generates around 250,000 units of electricity per year. For commercial applications, per unit cost of electricity is at least ₹ 12; generator-driven electricity costs are even higher. This amounts to a minimum of ₹ 3 million annually.

"In addition, there is a constant requirement for diesel (or other fuel) throughout the year. It is safe to say that, under these assumptions, total annual maintenance cost exceeds ₹ 3 million. The bigger the generator capacity, the higher the cost.

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